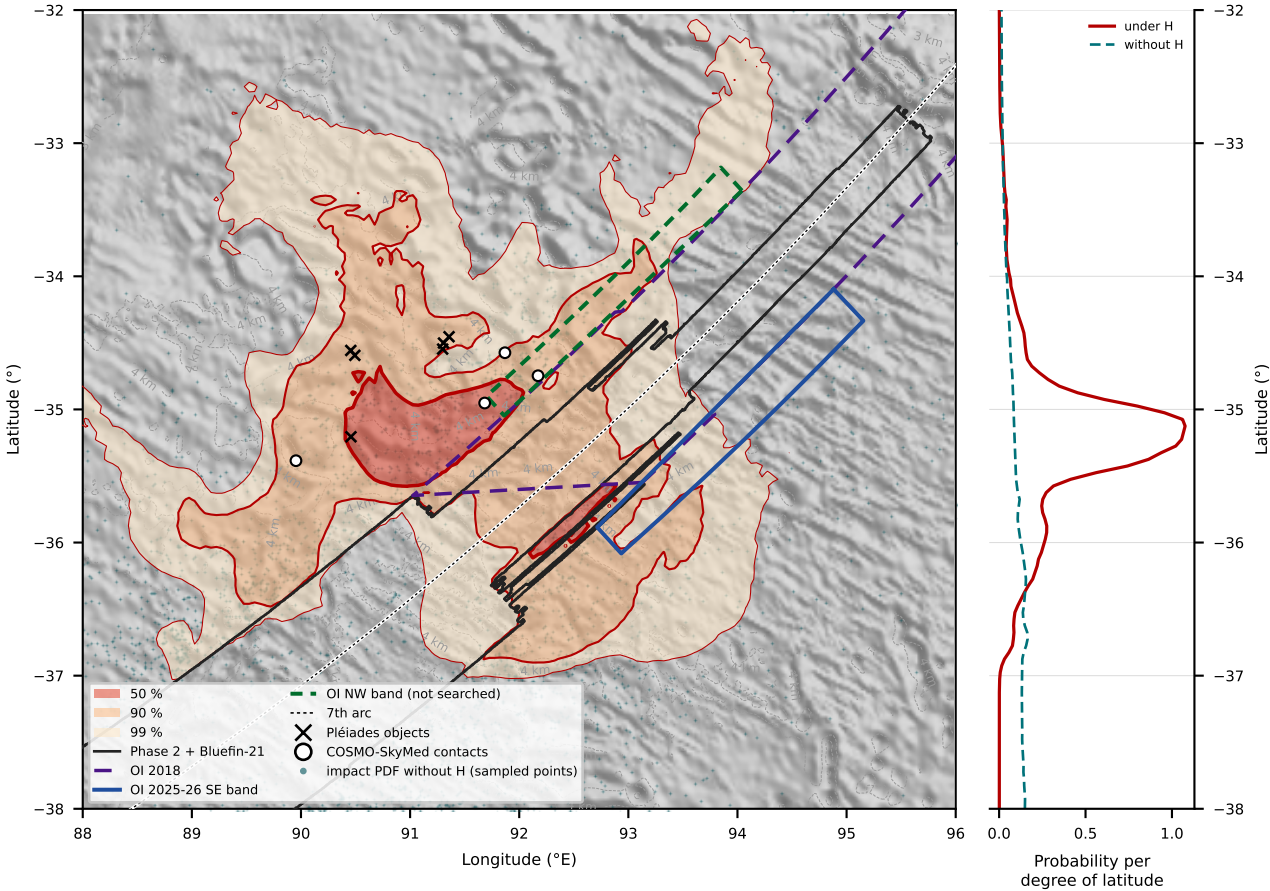


If the imaged objects are from 9M-MRO: where the debris entered the sea, on GEBCO 2026 relief — 00:19 Held Out

Pléiades + all four COSMO-SkyMed contacts, one debris field after Phase 2 + Bluefin-21 + OI 2018 + 2025-26 (grade C)



Source: impacts next-run (strata next-free, next-repro-radar, next-descent-climb, next-routes; provenance read from next-free) (4 seeds pooled with equal weight; source run src-next-free, code 3c6319f); prior track 289.7 deg (sd 1.0); config chain (run.json): davey2016-inmarsat + no-exhaustion-prior + s1-factor + s2-temperature + s3-internal + s4-ceiling + s5-hard-reject + reference-snapshots + radar-full + free + server-free + hypotheses/end-of-flight/smoke/snapshot-m0011.toml + hypotheses/end-of-flight/smoke/terminal.toml + hypotheses/end-of-flight/full/reference-289.toml + hypotheses/end-of-flight/full/descent-idle-floor.toml + hypotheses/end-of-flight/full/seed-1.toml; tempered sampler (m1839, m1941, m2041, m2141, m2241, m0011, 16 stages); particles per mode [500000, 250000, 1250000, 250000, 1250000]; BFO bias (run.json bfo_bias): {"mean_hz":150.0,"sd_hz":25.0}.

What was run: 00:19 Held Out: none of the 00:19 BTO/BFO values scored (held out); log-on not from fuel exhaustion (no lag term); aircraft transmitting at 00:19:37 and not powered at 01:15:56 (the two observed facts; end of flight's 'unpowered' existence constraint, ruling ~19:10 UTC B (b)) [end-of-flight arm: none+unpowered]. Strata mixed by P(family) re-weighted by the 00:19 evidence (ruling C; end of flight's family-evidence-next-run-b.json, key '00:19 Held Out +alive': next-free 0.6946, next-repro-radar 0.1523, next-descent-climb 0.1383, next-routes 0.0148; the not-powered-at-01:15:56 factor acts within each stratum, not in these weights); the core's fixed P(family) (next-free 0.6948, next-repro-radar 0.1527, next-descent-climb 0.1376, next-routes 0.0149) is shown beside it. The search then re-weights strata by their own evidence.

Labels: core (b): request-17 sampler fix IN, split-half NOT converged; two-tank bookkeeping only; internal-v1 fuel (one-engine flow 2x its tables); Inmarsat ephemeris; descent idle floor ON; EoF sweep run by an architecture stand-in; PROVISIONAL-OVERNIGHT.

Search: the searched-areas module's own per-impact likelihood (point target, rho = 0.05, shared). Phase 2 (q 0.945), Bluefin-21 (q 0.9) official; OI 2018 (q 0.9, coverage 0.889) and OI 2025-26 south-east band (q 0.9, coverage 0.7808) are community tracings, grade C; the north-west band is drawn only, never negative evidence.

Under H: 12 rating-5 Pléiades objects in 6 clusters (3 km), equal weights; COSMO-SkyMed contacts equally weighted, dawn 20 / dusk 21 Mar passes equal (unverified source, time, footprint); 'one debris field' = likelihoods multiplied per ocean model; GLORYS12 + ERA5 and GlobCurrent daily + ERA5 equal weight; measured transport error (~100-110 km per component at 15 d); windage 0-5 %; Pléiades and COSMO errors independent (rho = 0).

Conditional on H (the objects are debris from 9M-MRO); no Bayes factor or P(H | data).

Strata weights, after all searches, 90 % region under H: re-weighted 71,925 km², mean 35.25 S 91.41 E; fixed 71,925 km², mean 35.25 S 91.41 E (drawn: re-weighted).

Tension of H against the flight PDF, after all searches (the module's measure; reported with the conditional): ln S +0.18, p 0.45 (Handley-Lemos), mean shift 97 NM; flight PDF mass inside the 90 % region under H 15 %.

Relief: GEBCO 2026 15-arcsecond grid (doi:10.5285/4f68d5c7-45eb-f999-e063-7086abc036fa), shown at 0.025 deg; contours every 1 km depth. Right: probability per degree of latitude over 85-103 E, after all searches.

Background: the impact PDF without H (flight posterior alone, after the same searches), as 6,000 points drawn in proportion to its probability from 0.05 deg grid cells and jittered within each cell. 45 % of it lies inside this frame.